

RF test equipment buyers in India

From private manufacturers who make amplifiers for cable TVs to large organisations like ISRO, DRDO, Intel, TI or STMicro, customers working on a variety of latest or cutting-edge technologies are buyers of RF test equipment. For example, ISRO and DRDO, who design satellite transponders and electronic warfare systems, respectively, require sophisticated and accurate instruments. Small- and mid-level manufacturers prefer cost-effective, multi-featured test equipment.

Pricing trends. Pricing trends vary depending on the market segment. For example, customers working on aerospace and defence-related applications or emerging wireless standards are ready to pay premium price for novel equipment. However, customers using general-purpose RF equipment are sensitive about their bottom-line performance. For such applications, prices of tools are decreasing. There is a higher requirement for cost-effective test solutions.

—Vishal Gupta, senior application consultant (RF/MW, Surveillance), Keysight Technologies



Fig. 6: HL2202 high-speed signal path analyser, which costs US\$ 7495 (Image courtesy: www.hyperlabsinc.com)

The simple and intuitive UI allows engineers to generate and complete the test report on-site, also allowing them to Geotag each test point and place a snapshot of Google Maps directly into the report.

Analysing signals in RF designs in a budget-friendly manner

Engineers typically measure the performance of RF and microwave devices by considering S parameters, or, in other words, network scattering parameters. A vector network analyser (VNA) is usually employed to ensure that the RF design of the circuit is developed to present the best performance. It would be safe to assume that VNAs are crucial for RF design engineers, but there is a new, budget-friendly device in the market that looks to replace VNAs for most common applications.

Engineers do not make use of the entire spectrum of a VNA for common signal verification applications including cable testing, which are usually expensive. In such cases, relevant test can be performed on HyperLab's signal path analysers (SPAs). This instrument measures the vital parameters influencing the course of a signal through PCB trace, high-speed interconnect or cables.

S parameters, namely, return loss (S11) and insertion loss (S21) measurement capabilities, are incorporated in the recently-launched

versions, eliminating the need for a separate network analyser, which is not to be seen in the low-cost competing instruments.

Limited to 20 channels, this tool can acquire and analyse cross-talk, time-domain transmissometry (TDT) and time-domain reflectometry (TDR) data in systems. Using the software, an engineer can also obtain voltage, impedance and other normalised readings in time-domain and frequency-domain. See Fig. 7 to see a comparison in the performance of a VNA and an SPA.

Spotting and fixing signal interference issues

The increasing use of Centralised Radio Access Network (C-RAN) architecture, macro cells, distributed antenna systems and small cells is propelling installation of fibre based remote radio units to attain high bandwidth and reduced inactivity.

Common Public Radio Interface (CPRI) and Open Base Station Architecture Initiative (OBSAI) protocols are extensively used in such environments. It has become even more important for technicians and engineers to identify and resolve signal interference issues swiftly not only from conventional mobile sites, but even modern sites that employ

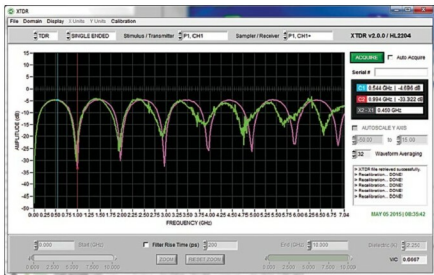


Fig. 7: Comparison of S11 (return loss) measurements taken by XTDR and HL2204 (green trace) versus those taken on a much more expensive VNA instrument (pink trace) (Image courtesy: www.everythingt.com)